Total No. of Questions : 8]

SEAT No. :

PD4105

[Total No. of Pages : 2

[6402]-65

S.E. (AI&ML)

DATA STRUCTURES & ALGORITHMS

(2019 Pattern) (Semester - III) (218542)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.***
2) ***Neat diagrams must be drawn wherever necesssary.***
3) ***Figures to the right indicate full marks.***
4) ***Assume suitable data if necessary.***

Q1) a) Write sudo code to read a valid postfix expression and evaluate the same. **[6]**

b) Enlist the applications of double ended queue & priority queue. **[6]**

c) Establish queue over flow and under flow conditions for linear, when sequential organization is used with max elements N. **[6]**

OR

Q2) a) Clearly indicate the content of stack during conversion of infix expression to prefix expression. **[9]**

i) (A + B) $ (D – E) / (F + G / I * 2)

ii) 5/× $ y – z * (P + Q)

b) Write sudo code for insert & delete operations for circular using when sequential organization is used for implementation. **[9]**

Q3) a) Write sudo code for creating an expression tree and Discuss its time complexity. **[8]**

b) Discuss with the help of example how threaded binary search trees, make the BST operations faster? **[6]**

c) Discuss the procedure of adding a node in a BST using example. **[4]**

OR

Q4) a) Write sudo code for inorder traversal technique for inorder - threaded binary tree. **[6]**

b) Construct an expression tree for the given postfix expression. Demonstrate the steps property. AB + CD – * **[6]**

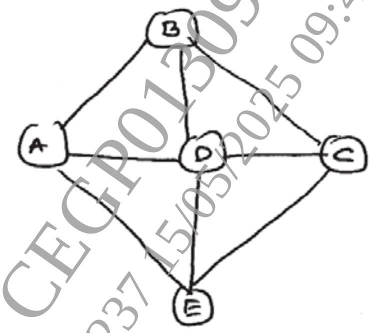
c) Discuss the time complexity of finding height and depth of a tree. **[6]**

P.T.O.

Q5) a) What is a minimum spanning tree. Discuss the various methods available for finding it. **[6]**

b) Enlist applications of graphs. Explain now the Dijkstra algorithm is contributing to computer networks. **[5]**

c) Discuss briefly the graph traversals & perform those traversals for the following graph. **[6]**



OR

Q6) a) Discuss how AVL is different from BST with suitable examples & illustrations? **[8]**

b) Enlist the characteristics of Heap data-structure. Give examples for max-heap & min-heap. **[6]**

c) Discuss the time complexities of BFS & DFS graph traversals. **[3]**

Q7) a) Discuss the significance of the following. Illustrate your explaination with the help of examples. **[9]**

i) Primary indexes

ii) Clustering indexes

iii) Secondary indexes

b) Explain the concept of hashing. How hashing impacts the search time. Justify your answer with time complexity. **[8]**

OR

Q8) a) Discuss the types of collision resolution techniques for hash - tables. **[8]**

b) Enlist the file - opening modes in C++ and discuss any three briefly. **[9]**

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